Gauge Theories

Nishtha Tikalal

Contents

1 Gauge Theories (Abelian 1

2 Non–Abelian Gauge Theories 3

3 Higgs Fields 4

1. Gauge Theories (Abelian

We begin by recalling Maxwell's equations:

$$\text{div } \boldsymbol{B} = 0 \tag{1}$$

$$\frac{\partial \boldsymbol{B}}{\partial t} + \text{curl } \boldsymbol{E} = 0 \tag{2}$$

$$\text{div } \boldsymbol{E} = \rho \tag{3}$$

$$\frac{\partial \boldsymbol{E}}{\partial t} - \text{curl } \boldsymbol{E} = -\boldsymbol{j}. \tag{4}$$

Also note that the electric and magnetic fields can be expressed using a vector potential:

$$\boldsymbol{B} = \text{curl } \boldsymbol{A}$$

$$\boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t} - \text{grad } \phi.$$

Maxwell's equations are invariant under the gauge transformation

$$A_\mu \to A_\mu + \partial_\mu \chi \tag{1.243}$$

where χ is a scalar function. This invariance is manifest if we define the **electromagnetic field tensor** $F_{\mu\nu}$ by

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_z \\ E_z & B_y & -B_x & 0 \end{pmatrix}. \tag{1.244}$$

From the construction, F is invariant under (1.243). The Lagrangian of the electromagnetic fields is given by

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu j^\mu \tag{1.245}$$

where $j^\mu = (\rho, \boldsymbol{j})$.

Consider a Dirac field ψ carrying an electric charge e. The baseline free Dirac Lagrangian is given by:

$$\mathcal{L}_0 = \bar{\psi}(i\gamma^\mu\partial_\mu + m)\psi \tag{1.247}$$

This free Lagrangian possesses a global symmetry ($\alpha \in \mathbb{R}$ is a constant) under the phase shift:

$$\psi \to e^{-ie\alpha}\psi, \quad \bar{\psi} \to \bar{\psi}e^{ie\alpha} \tag{1.248}$$

To promote this to a local gauge symmetry, we make the phase parameter space-time dependent so that $\alpha = \alpha(x)$:

$$\psi \to e^{-ie\alpha(x)}\psi, \quad \bar{\psi} \to \bar{\psi}e^{ie\alpha(x)} \tag{1.249}$$

Because the derivative acts on the space-time dependent phase, the free Lagrangian is no longer invariant under local transformations. Instead, it picks up an unwanted extra term:

$$\bar{\psi}(i\gamma^\mu\partial_\mu + m)\psi \to \bar{\psi}(i\gamma^\mu\partial_\mu + e\gamma^\mu\partial_\mu\alpha + m)\psi \tag{1.250}$$

To absorb this extra term and restore symmetry, we introduce a gauge vector field A_μ that couples directly to the Dirac field. This yields the interacting Lagrangian:

$$\mathcal{L} = \bar{\psi}[i\gamma^\mu(\partial_\mu - ieA_\mu) + m]\psi \tag{1.251}$$

This complete Lagrangian successfully preserves local gauge invariance, provided that both the Dirac field and the gauge field transform simultaneously according to:

$$\begin{aligned} \psi &\to \psi' = e^{-ie\alpha(x)}\psi \\ \bar{\psi} &\to \bar{\psi}' = \bar{\psi}e^{ie\alpha(x)} \\ A_\mu &\to A'_\mu = A_\mu - \partial_\mu\alpha(x) \end{aligned} \tag{1.252}$$

To simplify the notation for this gauge-invariant coupling, we define the covariant derivatives as:

$$\nabla_\mu \equiv \partial_\mu - ieA_\mu, \quad \nabla'_\mu \equiv \partial_\mu - ieA'_\mu \tag{1.253}$$

Under a local gauge transformation, the covariant derivative of the field, $\nabla_\mu\psi$, transforms covariantly (meaning it picks up the exact same phase factor as the field ψ itself):

$$\nabla'_\mu\psi' = e^{-ie\alpha(x)}\nabla_\mu\psi \tag{1.244}$$

Verification of Covariant Transformation

To explicitly verify this relation, we use the transformation rules established previously:

$$\begin{aligned} \psi' &= e^{-ie\alpha(x)}\psi \\ A'_\mu &= A_\mu - \partial_\mu\alpha(x) \\ \nabla'_\mu &= \partial_\mu - ieA'_\mu \end{aligned}$$

Let us apply the transformed covariant derivative ∇'_μ to the transformed field ψ':

$$\nabla'_\mu \psi' = \left(\partial_\mu - ieA'_\mu\right)\left(e^{-ie\alpha(x)}\psi\right)$$

Expanding this expression gives two parts:

$$\nabla'_\mu \psi' = \partial_\mu\left(e^{-ie\alpha(x)}\psi\right) - ieA'_\mu\left(e^{-ie\alpha(x)}\psi\right)$$

First, apply the product rule to the partial derivative term:

$$\partial_\mu\left(e^{-ie\alpha(x)}\psi\right) = \left(-ie\partial_\mu\alpha(x)\right)e^{-ie\alpha(x)}\psi + e^{-ie\alpha(x)}\partial_\mu\psi$$

Next, substitute the explicit form of A'_μ into the second term:

$$-ieA'_\mu\left(e^{-ie\alpha(x)}\psi\right) = -ie\left(A_\mu - \partial_\mu\alpha(x)\right)e^{-ie\alpha(x)}\psi$$

$$= -ieA_\mu e^{-ie\alpha(x)}\psi + ie\left(\partial_\mu\alpha(x)\right)e^{-ie\alpha(x)}\psi$$

Now, combine both expanded terms back together:

$$\nabla'_\mu \psi' = \underbrace{\left(-ie\partial_\mu\alpha(x)\right)e^{-ie\alpha(x)}\psi}_{\text{Term 1}} + e^{-ie\alpha(x)}\partial_\mu\psi - ieA_\mu e^{-ie\alpha(x)}\psi + \underbrace{ie\left(\partial_\mu\alpha(x)\right)e^{-ie\alpha(x)}\psi}_{\text{Term 2}}$$

Notice that **Term 1** and **Term 2** cancel each other out perfectly. This leaves:

$$\nabla'_\mu \psi' = e^{-ie\alpha(x)}\partial_\mu\psi - ieA_\mu e^{-ie\alpha(x)}\psi$$

Factoring out the shared phase factor $e^{-ie\alpha(x)}$ yields:

$$\nabla'_\mu \psi' = e^{-ie\alpha(x)}\left(\partial_\mu - ieA_\mu\right)\psi$$

Substituting the definition of the original covariant derivative $\nabla_\mu = \partial_\mu - ieA_\mu$ completes the proof:

$$\nabla'_\mu \psi' = e^{-ie\alpha(x)}\nabla_\mu\psi \tag{5}$$

By combining the gauge-invariant kinetic term for the electromagnetic field tensor $F_{\mu\nu}$ with the locally symmetric Dirac field term, we arrive at the complete **Quantum Electrodynamic (QED) Lagrangian**:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \bar{\psi}(i\gamma^\mu\nabla_\mu + m)\psi \tag{6}$$

2. Non–Abelian Gauge Theories

The standard $U(1)$ gauge transformation is Abelian because its elements are complex phases (modulus 1), which commute. Moving beyond this, Yang and Mills introduced non-Abelian gauge theories, where the transformation elements do not commute. These non-Abelian theories form the mathematical bedrock of modern elementary particle physics.

Let G represent a compact semi-simple Lie group (common physical examples include $SO(N)$ or $SU(N)$). The anti-Hermitian generators of the group, denoted by $\{T_\alpha\}$, satisfy the fundamental commutation relations:

$$[T_\alpha, T_\beta] = f_{\alpha\beta}{}^{\gamma}T_\gamma$$

where the coefficients $f_{\alpha\beta}{}^{\gamma}$ are the **structure constants** characterizing the specific Lie algebra of G.

An infinitesimal group element $U \in G$ close to the identity can be expressed exponentially via these generators:

$$U = \exp(-\theta^{\alpha} T_{\alpha})$$

We consider a Dirac field ψ transforming under the group element $U \in G$ (typically assumed to belong to the fundamental representation of the group):

$$\psi \to U\psi, \quad \bar{\psi} \to \bar{\psi}U^{\dagger}$$

To build a gauge-invariant theory under this non-Abelian group, we consider an interacting Lagrangian of the form:

$$\mathcal{L} = \bar{\psi}[i\gamma^{\mu}(\partial_{\mu} + gA_{\mu}) + m]\psi$$

where g is the coupling constant, and A_{μ} represents the non-Abelian gauge field.

The Yang–Mills gauge field A_{μ} is Lie-algebra-valued, meaning it can be decomposed as a linear combination of the anti-Hermitian generators: $A_{\mu} = A_{\mu}^{\alpha} T_{\alpha}$. The coupling constant g determines the strength of the interaction between the matter field and this gauge field. The entire Lagrangian remains invariant under the local gauge transformations:

$$\begin{aligned} \psi &\to \psi' = U\psi \\ \bar{\psi} &\to \bar{\psi}' = \bar{\psi}U^{\dagger} \\ A_{\mu} &\to A'_{\mu} = UA_{\mu}U^{\dagger} + g^{-1}U\partial_{\mu}U^{\dagger} \end{aligned}$$

Using the same logic as the Abelian case, we establish the non-Abelian covariant derivative as $\nabla_{\mu} = \partial_{\mu} + gA_{\mu}$. This operator ensures that the derivative of the field transforms nicely under a gauge transformation by picking up the group element directly:

$$\nabla'_{\mu}\psi' = U\nabla_{\mu}\psi$$

Because the field components do not commute, the non-Abelian field strength tensor (the Yang–Mills field tensor) includes an extra commutator term:

$$\mathcal{F}_{\mu\nu} \equiv \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu} + g[A_{\mu}, A_{\nu}]$$

By projecting this tensor onto the algebra generators using the structure constants $f_{\beta\gamma}{}^{\alpha}$, we find its individual component form to be:

$$F_{\mu\nu}{}^{\alpha} = \partial_{\mu}A_{\nu}^{\alpha} - \partial_{\nu}A_{\mu}^{\alpha} + gf_{\beta\gamma}{}^{\alpha}A_{\mu}^{\beta}A_{\nu}^{\gamma}$$

Furthermore, if we construct the dual field tensor as $*\mathcal{F}^{\mu\nu} \equiv \frac{1}{2}\varepsilon^{\mu\nu\kappa\lambda}\mathcal{F}_{\kappa\lambda}$, it satisfies a generalized, gauge-covariant Bianchi identity:

$$\mathcal{D}_{\mu} * \mathcal{F}^{\mu\nu} \equiv \partial_{\mu} * \mathcal{F}^{\mu\nu} + g[A_{\mu}, *\mathcal{F}^{\mu\nu}] = 0$$

3. Higgs Fields

We observe that most vector gauge fields are massive in nature, except for the photon. This physical reality requires us to break gauge symmetry. To ensure the theory remains renormalizable, this symmetry breaking must happen spontaneously via the Higgs mechanism.

To model this, we consider an Abelian $U(1)$ gauge field coupled to a complex scalar field ϕ. The system is described by the Lagrangian:

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + (\nabla_\mu\phi)^\dagger(\nabla^\mu\phi) - \lambda(\phi^\dagger\phi - v^2)^2$$

The scalar potential $V(\phi) = \lambda(\phi^\dagger\phi - v^2)^2$ has a non-trivial set of minima where $V = 0$ at the radius $|\phi| = v$. The original Lagrangian remains invariant under standard local transformations:

$$\begin{aligned} A_\mu &\to A_\mu - \partial_\mu\alpha \\ \phi &\to e^{-ie\alpha}\phi \\ \phi^\dagger &\to e^{ie\alpha}\phi^\dagger \end{aligned}$$

When the system picks a specific vacuum state, the scalar field develops a non-zero vacuum expectation value (VEV) $\langle\phi\rangle$, which spontaneously breaks the $U(1)$ symmetry. We can parameterize fluctuations around this vacuum state by expanding ϕ in terms of its radial component $\rho(x)$ and phase component $\alpha(x)$:

$$\phi = \frac{1}{\sqrt{2}}[v + \rho(x)]e^{i\alpha(x)/v} \sim \frac{1}{\sqrt{2}}[v + \rho(x) + i\alpha(x)]$$

Assuming a non-vanishing vacuum value $v \neq 0$, we can choose the unitary gauge to effectively absorb the phase component $\alpha(x)$ into a redefined gauge field. This 'gauges away' the phase, leaving only the real part of the scalar field:

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \rho(x))$$

Substituting this unitary gauge parameterization back into the original Lagrangian and expanding in terms of the physical fluctuation field ρ gives:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu\rho\partial^\mu\rho + \frac{1}{2}e^2A_\mu A^\mu(v^2 + 2v\rho + \rho^2) - \frac{1}{2}\lambda(4v^2\rho^2 + 4v\rho^3 + \rho^4)$$

From the uncoupled, free parts of this expanded Lagrangian, we derive the classical equations of motion for both fields:

$$\partial^\nu F_{\nu\mu} + 2e^2v^2A_\mu = 0, \quad \partial_\mu\partial^\mu\rho + 2\lambda v^2\rho = 0$$

Applying the divergence to the field equation for A_μ forces the gauge field to obey the Lorentz condition $\partial_\mu A^\mu = 0$. In terms of physical degrees of freedom, the system keeps a constant total count of 4: it shifts from having 2 massless photon polarizations plus 2 complex scalar modes into having 3 massive vector boson components plus 1 real scalar particle (ρ). Note that the temporal component A_0 develops a mass term with an unphysical sign, confirming it does not represent an independent physical degree of freedom. This entire process of a gauge boson absorbing a Goldstone mode to gain mass is called the Higgs mechanism.

References

[1] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. Bristol, UK: Institute of Physics Publishing, 2003.